Development of Nano-structured $TiO_2/CaCO_3$ Core Shell Materials for Efficient Dye-sensitized Solar Cells

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***Abstract*—** $TiO_2/CaCO_3$ core/shell nanomaterials were synthesized by simple solution growth technique. TiO_2 nanoparticles were first coated on fluorine doped tin oxide and then $CaCO_3$ shell layer was deposited on TiO_2 nanoporous thin film to form core/shell structure. The different thickness of calcium carbonate layer and illumination area of photoelectrodes (i.e. 0.25 cm^2 and 1cm^2) could affect the photovoltaic performance of the dye sensitized solar cells. The present work clearly reveals that the dye-sensitized solar cells based on the $TiO_2/CaCO_3$ core/shell nanomaterials are superior compared to the bare TiO_2 based counter-parts. The overall conversion efficiency of dye-sensitized solar cell was enhanced from 8.1 to 9.1%.

Keywords—$TiO_2/CaCO_3$; Core/Shell; Nanomaterials; Dye-sensitized Solar Cells

I. INTRODUCTION

Dye sensitized solar cells (DSCs) are promising and alternative to conventional silicon solar cells due to their low cost production and easy manufacturing process [1-5]. Various materials have been developed to fabricate the highly efficient DSCs such as SnO_2, ZnO, MgO, Nb_2O_5 and TiO_2. TiO_2 has a peculiar and fascinating physicochemical property and it is used in various applications such as dye sensitized solar cells, photocatalysis, gas sensors and supercapacitor etc., [6-18]. The DSCs consists of a TiO_2 nanocrystalline layer deposited on a transparent conducting oxide (TCO) which is used as a photoanode, a ruthenium complex dye, an electrolyte solution containing an iodide/tri-iodide redox couple and a Pt- counter electrode. The sensitizer molecule absorbs light photons from sunlight and excites the electron from HOMO level to LUMO. Then, the photo excited electrons are injected from the LUMO of dye into the conduction band of semiconductor and collected at anode. The electrons can be regenerated by electrolyte solution and the holes are transported to the cathode where the Pt layer acts as catalyst for reduction of tri-iodide [19].

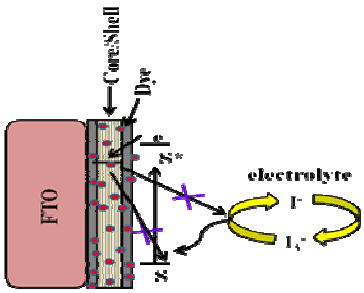

Fig.1. Schematic approach to suppress the interfacial recombination of core/shell structured electrode

The conversion efficiency of DSCs can be improved by coating insulating layer which will reduce the recombination of electrons from conduction band of TiO_2 to the electrolyte solution. The insulating nature of coating layer can increase the surface resistance of TiO_2 nanoparticles [20]. The back transfer of electrons can be reduced using a new alternative core/shell structured nanomaterials. The nanomaterial gives different morphologies such as nanoparticles, nanowires, nanorods, nanosheet etc., [4, 21]. The performance of photoanode material depends on several factors like morphology, pore volume and crystallinity of the nanomaterials. All these factors influence the charge transport and recombination process and ultimately control light to electricity conversion efficiency of DSCs. The core/shell structured materials play an important role for improving the conversion efficiency of the cell [20, 22]. The interfacial recombination of injected electrons from photo excited dye to the dye cations and to the redox couple limiting the efficiency of DSCs [23, 24]. The growth of insulating thin layers on the surface of nanoporous films may be important approach to suppress the interfacial recombination as shown in schematic representation (Fig. 1) [25, 26]. Here, in the present work $CaCO_3$ has been selected as the coating layer, because $CaCO_3$ has high band gap (6.0eV) and its isoelectric point (IEP) is 8.3. To physically separate the injected electrons and electron acceptors and retard charge recombination in DSCs to helpful nature of $CaCO_3$, it is an electronic insulator. The alkaline nature of $CaCO_3$ favor higher dye adsorption and thus higher light harvesting efficiency (LHE) in the DSCs. In this study, $TiO_2/CaCO_3$ has been synthesized by solution growth technique. The properties of electron transport and recombina-

tion in the TiO_2 nanoporous films with different thickness of calcium carbonate was also investigated to optimize the photovoltaic performance of DSCs. Morphological, structural and optical properties of the synthesized nanoparticles have been studied and its successful use in DSCs has also been reported.

II. EXPERIMENTAL DETAILS

A. Materials and methods

The $CaCO_3$-coated TiO_2 electrodes were prepared by solution growth technique. The Fluorine doped Tin Oxide (FTO) glass was used as the substrate. The substrate was ultrasonically cleaned with acetone. Then, the TiO_2 colloidal solution was coated on FTO glass substrate by doctor-blade technique. The surface profilometer was used to determine the film thickness, surface roughness and topography of the prepared thin films. The thickness of the coated $CaCO_3$ was calculated using X-ray Photoelectron Spectroscopy (XPS). The morphology of the films was observed by Scanning Electron Microscopy (SEM). The Pt was used as counter electrode. The mixture of (0.6 M dimethyl imidazolium iodide + 0.1 M LiI + 0.05 MI_2 + 0.5 M t-butyl pyridine in methoxy acetonitrile) Tetrabutylammoniumcis-di(thiocyanato)-N,N'-bis-(4carboxylato-4'-carboxylic acid-2,2'-bipyridine)ruthenate(II) (N719) were procured from solaronix. EIEKO Model MP-160S IV–tracer was used to calculate the photocurrent-voltage of DSCs and Wacom solar simulator equipped with AM 1.5 filter was utilized as a light source.

B. Preparation of TiO_2 and $CaCO_3$ coated TiO_2 Electrodes

The bare TiO_2 and $CaCO_3$- coated TiO_2 films were prepared by following procedure. The nitric acid-treated TiO_2 (P-25) powder was mixed with water and homogenized for 10 min using ultrasonic homogenizer [9]. Finally, [Ti(OH)x(OOH)y] and Triton X-100 were added into the homogenizer and stirred for 1-2 hrs using magnetic stirrer. Then, the paste was deposited on FTO glass and heated at 500°C for 30 min. In addition, various thick $CaCO_3$ layer containing TiO_2 films were prepared by dipping the bare TiO_2 film into a solution of various weight % of calcium acetate dissolved in water (various time interval). Finally, $CaCO_3$ shell layer on TiO_2 surface was obtained when it was calcined at 500°C for 30 min.

C. Solar Cell Construction and Photoelectrochemical Measurements

All the prepared samples were immersed into the 0.3 M ethanolic solution of N719, while still it is in hot condition for 1 hour at 80°C. The dye adsorption takes place on $TiO_2/CaCO_3$ films while heating and subsequently dye modified $TiO_2/CaCO_3$ films were stored at N2 atmosphere. Dye-modified films were rinsed with acetonitrile, dried at room temperature and then used for photoelectrochemical measurements. UV-visible absorption spectra were taken for all the dye-modified thin films to measure the amount of dye loaded on the $TiO_2/CaCO_3$ films. For the I-V measurements, the dye-modified $TiO_2/CaCO_3$ films and Pt coated conducting glass were used as the working electrode and cathode, respectively. A drop of 0.6 M dimethyl imidazolium iodide + 0.1 M LiI + 0.05 MI2 + 0.5 M t-butyl pyridine in methoxy acetonitrile containing liquid mixture was used as electrolyte. The cells were directly tested under AM 95mW/cm^2 simulated sunlight. The schematic structure of constructed DSCs is shown in Fig. 2.

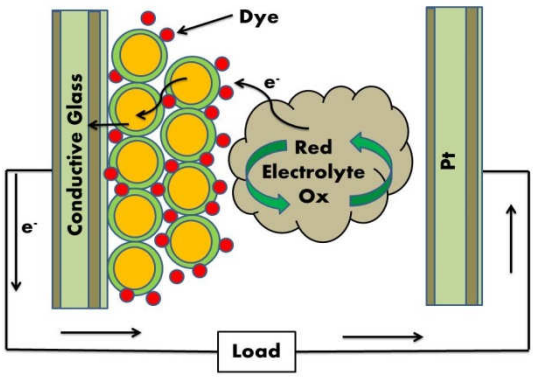

Fig.2. Schematic structure of constructed dye-sensitized solar cells using $TiO_2/CaCO_3$ core/shell structured electrode

III. RESULTS AND DISCUSSION

The $CaCO_3$ coated on TiO_2 were synthesized by hydrolyzing $Ca(CH_3COO)_2$, which was followed by heat-treatment in air. The aqueous solution of $Ca(CH_3COO)_2$, Ca^{2+} ions could be adsorbed particularly onto the TiO_2 surface through exchange with protons at the oxide/water interface. The reaction routes are described as follows;

$$Ca(CH_3COO)_2 + H_2O \longrightarrow Ca(OH)_2 + CH_3COOH$$

$$Ca(OH)_2 - CO_3^{2-} \longrightarrow Ca(CO)_3 + 2OH^-$$

It is evidently known that $Ca(OH)_2$ transforms into $CaCO_3$ in case that it is calcined in an atmosphere with carbon sources, e.g., CO_2 in air [7, 27].

The mean thickness of $CaCO_3$ on TiO_2 films were calculated from XPS analysis. The thickness of bare TiO_2 was about 8μm and the coated $CaCO_3$ was calculated as 0.14 nm from the XPS. Fig. 3 shows the scanning electron microscopic image of TiO_2 thin film and it confirms the formation of homogeneous morphology. The pore size of TiO_2 was found in the range of 50 nm.

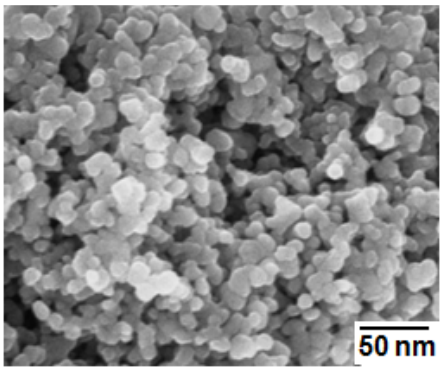

Fig.3. Scanning Electron Microscopy (SEM) image of TiO_2 thin film

To verify the formation of $CaCO_3$ on TiO_2, we prepared thick layer of $CaCO_3$ by adding 10 wt % of calcium acetate. The amount of $CaCO_3$ loaded on TiO_2 film (Ca/Ti) and average thickness of $CaCO_3$ was calculated. From XPS, we assumed that all the added calcium acetate was converted into $CaCO_3$ and deposited uniformly on TiO_2 owing to the specific adsorption of Ca^{2+} for thickness calculation.

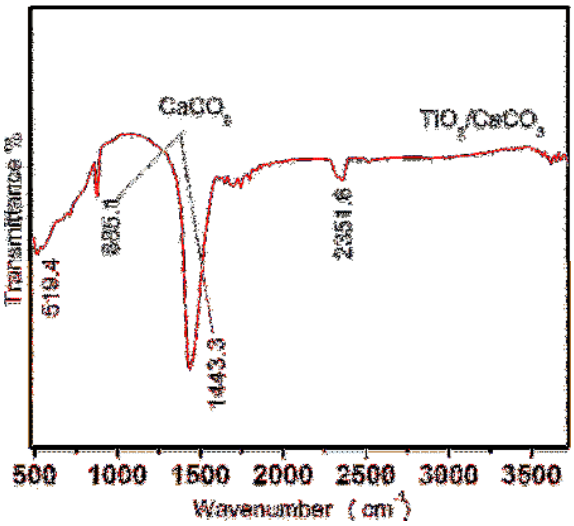

Fig.4. FTIR spectra of $TiO_2/CaCO_3$

FT-IR spectrum were confirms the presence of $CaCO_3$ which were shown in Fig. 4. The peaks observed at 400-900 cm^{-1} indicates the stretching vibration of inorganic Ti-O in the spectrum of $TiO_2/CaCO_3$. The FT-IR spectrum reveals the presence of C-O in CO_3^{2-} in the range of 1443 cm^{-1} due to the antisymmetric stretching vibration as well as the band at 885 cm^{-1} due to the plane of bending vibration. This result strongly supports the development of carbonate upon calcium acetate treatment followed by calcinations. The peak appeared at 2351 cm^{-1} and 3444 cm^{-1} may be due to the adsorption of water molecules from the moisture [28-31].

Fig. 5 (a) shows typical ultraviolet-visible (UV-vis) absorption spectra of TiO_2 and $TiO_2/CaCO_3$ films. The optical band gap (Eg) of TiO_2 and $TiO_2/CaCO_3$ can be determined from the Tauc plot of the absorption spectra and as shown in the Fig. 5 (b). As expected, the TiO_2 film exhibits a light absorption onset at 398 nm. After the deposition of $CaCO_3$, besides the light absorption onset at 398 nm, another light absorption onset in the visible light region at wavelength of about 413 nm was also observed. Furthermore, it was observed that the absorbance in visible light region increases with the increasing concentration of calcium acetate solution. The band gap of TiO_2 and $TiO_2/CaCO_3$ has been found to be 3.05 eV and 2.92 eV, respectively.

The conversion efficiency of DSCs is mainly depends on four factors i.e. short circuit current, open-circuit voltage, fill factor and intensity of photo-incident light. The efficiency can be calculated according to the following equation

$$\eta = \frac{Jsc\ Voc\ FF}{P_{in}} \times 100 \qquad (1)$$

where, the conversion efficiency called as η, Jsc- short circuit current, Voc- open circuit voltage, FF- fill factor and and P_{in} -is the intensity of the incident light.

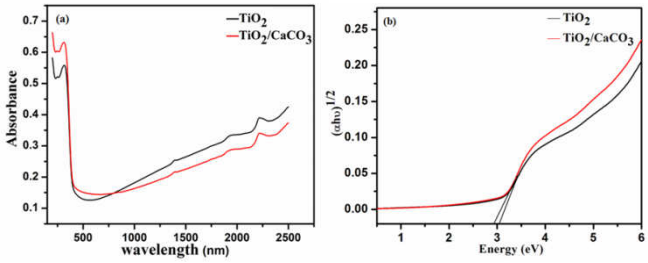

Fig.5. UV- visible absorption spectra (a) and optical absorption edges (b)

The fill factor is derived from I-V curve

$$FF = \frac{}{I_m \times V_m} \qquad (2)$$

where, V_m- is the maximum output current, I_m- is the maximum output voltage.

The photoelectrochemical performances of DSCs were measured to investigate the effect of barrier layer thickness ($CaCO_3$) on TiO_2 films. The bare TiO_2 nanoporous electrode served as a reference. The other $CaCO_3$ coated TiO_2 electrodes were used for the analysis of their real performance in presence of $CaCO_3$ layer. To ensure fair comparison, all the cells were studied under the similar conditions and only varying the illumination area i.e. 0.25 cm^2 and 1cm^2 [32-36].

The conversion efficiency can be depends on open-circuit voltage (Voc), short circuit current (Jsc), fill factor (FF), efficiency (η) and dye concentration. The photoelectrochemical performance of bare TiO_2 based DSCs which exhibits an open-circuit voltage (Voc) of 674, a Jsc of 16.56, a fill factor (FF) of 0.695 and an efficiency (η) of 8.1% whereas decorated $TiO_2/CaCO_3$ core/shell shows a Voc of 715, a Jsc of 17.52, a fill factor (FF) of 0.696 and efficiency (η) of 9.1% (Table 1).

TABLE 1. COMPARISON OF PHOTOELECTROCHEMICAL PERFORMANCE PARAMETERS OF THE CELLS BASED ON TiO_2 AND $TiO_2/CaCO_3$ FILMS, (LIGHT INTENSITY; 95 mW/cm^2. ILLUMINATION AREA; 0.25 cm^2)

Materials	Jsc (mA/cm^2)	Voc (mV)	F.F.	Efficiency %
TiO_2	16.56	674	0.695	8.1
$TiO_2/CaCO_3$	17.52	715	0.696	9.1

The reduced interfacial recombination may be responsible for the significant increase of Voc and fill factor as reported earlier [35]. The coating of $CaCO_3$ layer behaves as an energy barrier to separate the injected electrons from oxidized dye cations and I_3^- ions physically and thus increases the electron life time. The results indicate the increase of Voc due to the increased life time of the electrons [37].The better dye adsorption on $TiO_2/CaCO_3$ surface may be due to the more basic nature of $CaCO_3$, which favors the attachment of the acidic dye molecules through its anchoring carboxylic acid groups [35]. The back transfer of electrons from $CaCO_3$-coated TiO_2 nanoporous photoelectrodes to the electrolyte at the TiO_2/dye/electrolyte interface may also be prevented [20].

In order to further investigate the effect of cell area on the performance of DSCs, several cells were fabricated with various cell areas (Table 1 and Table 2). The efficiency of the DSCs made of $TiO_2/CaCO_3$ core/shell structured material was gradually decreases with increasing concentration of calcium acetate solution due to the formation of thick layer of $CaCO_3$ on TiO_2 surface as shown in Table 2.

The sharp decreases in the Jsc value may be due to the reduction of tunnelling probability of electrons across the thicker $CaCO_3$ barrier layer (Fig 6 (a)). The cell parameters such as Voc and FF are gradually increases with increasing concentration of calcium acetate solution (Fig. 6 (b) and (c)). The thick layer of outer shell may protect the electron injection on the TiO2 surface from dye and thus gives a poor performance of the cell.

TABLE 2. COMPARISON OF PHOTO ELECTROCHEMICAL PERFORMANCE PARAMETERS OF THE CELLS BASED ON TiO_2 AND $TiO_2/CaCO_3$ FILMS, (LIGHT INTENSITY; 95 mW/cm^2. ILLUMINATION AREA; 1 cm^2)

Calcium acetate Conc. (M)	Jsc (mA/cm^2)	Voc (mV)	F.F.	Efficiency %	Dye Concentration (10^{-8} mol/cm^2)
0	12.57	695.304	0.391	3.576	4.4
0.04	9.464	760.164	0.52	3.914	4.1
0.08	9.613	743.265	0.54	4.037	4.5
0.15	9.731	762.112	0.556	4.317	4.5
0.2	9.79	770.283	0.556	4.386	5.7
0.25	9.205	739.758	0.463	3.301	5.6

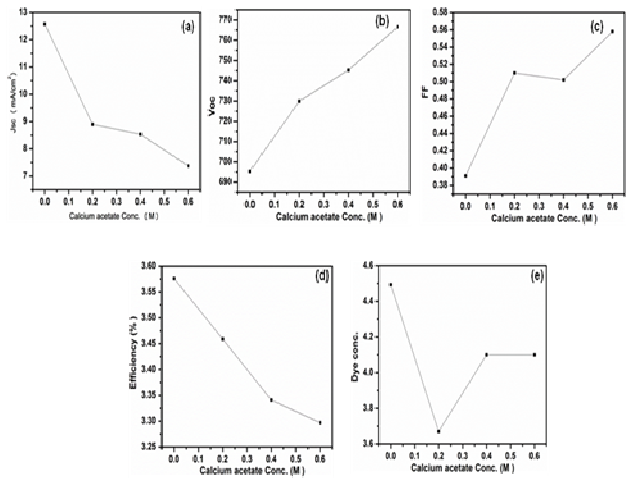

Fig.6. The performance of DSCs made of $TiO_2/CaCO_3$ core/shell materials as a function of different mole concentration of calcium acetate at a light intensity of 95 mW/cm^2. This graph displays an effect of calcium acetate on (a) short circuit current (Jsc), (b) open circuit voltage (Voc), (c) fill factor (FF), (d) Efficiency (η) (e) dye concentration (Illumination area; 1 cm^2)

The prevention of back transfer of electrons may be responsible for the increase of Voc and Fill factor as reported earlier [21-23]. It is quite interesting to note that the improvement of Jsc is observed in the $TiO_2/CaCO_3$ electrode comprising cell despite of the fact that the existence of $CaCO_3$ layer tends to minimize the effectiveness of electron injection from excited dye molecules into the conduction band of TiO_2. The increases in the dye adsorption on $CaCO_3/TiO_2$ electrodes should be responsible for this higher photocurrent (Table 2). Table 2 show better dye adsorption on $CaCO_3/TiO_2$ surface and this may be due to the more basic nature of $CaCO_3$, which favors the attachment of the acidic dye molecules through its anchoring carboxylic acid groups. The cell constructed with the electrodes of thicker $CaCO_3$ layer show higher Voc and Fill Factor, which confirms the blocking nature of $CaCO_3$. The 0.2 M calcium acetate forms the suitable thickness containing $CaCO_3$ layer on TiO_2 and ultimately gives higher efficiency of the cell. Finally, we confirmed from the Table 1 and 2, the thick layer of the outer shell may protect the electron injec-

tion on the TiO_2 surface from dye and thus gives a poor efficiency of the cell.

IV. CONCLUSIONS

Core-shell structured $TiO_2/CaCO_3$ electrodes have been successfully synthesized by solution growth technique. The characterization of resulting electrodes was carried out by Scanning electron microscopy, X-ray photoelectron spectroscopy and IV measurements. The covering of insulating shell layer on the core semiconductor oxide material reduces the interfacial recombination in DSCs. The increases in solar cell performance parameters have been observed in $TiO_2/CaCO_3$ based DSCs. The loading of dye on TiO_2 was also improved after the coating of $CaCO_3$ thin layer. The $CaCO_3$ coating works as an energy barrier at the electrode-electrolyte interface. The conversion efficiency of 9.1% was achieved for $TiO_2/CaCO_3$ based DSCs.

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